

Table 1: Predictors of treatment adoption relative to no treatment: Households

Group	Predictor	Any treatment			Boiling			Chlorination/tablets			Straining/settling		
		LASSO (coef.)	RF (imp.)	XGB (imp.)	LASSO (coef.)	RF (imp.)	XGB (imp.)	LASSO (coef.)	RF (imp.)	XGB (imp.)	LASSO (coef.)	RF (imp.)	XGB (imp.)
<i>Household wealth</i>	Wealth-index quintile 2	+0.11	0.4	0.1	+0.24	0.5	0.1	-0.04	1.6	0.7	-0.02	0.2	0.1
	Wealth-index quintile 3	+0.30	0.3	0.1	+0.44	0.3	0.1	+0.04	1.3	0.5	+0.09	0.2	0.3
	Wealth-index quintile 4	+0.49	0.4	0.2	+0.53	0.4	0.2	+0.44	1.6	0.7	+0.05	0.2	0.2
	Wealth-index quintile 5	+1.00	1.1	0.4	+0.80	0.7	0.3	+0.80	3.5	2.3	+0.19	0.3	0.3
<i>Residence</i>	Urban residence	-0.03	0.8	0.2	-0.02	0.8	0.3	+0.07	3.2	0.7	-0.28	0.6	0.7
	Drinking-water source												
<i>Drinking-water source</i>	Other water source	+0.02	1.2	0.4	-0.12	2.5	0.0	-0.15	1.5	1.9	+0.19	0.0	0.2
	Packaged water	-2.71	5.9	3.8	-3.10	5.1	3.8	-2.62	9.9	12.2	-2.39	1.6	3.5
	Protected well or spring	+0.57	1.9	0.9	+0.35	2.9	1.7	+0.76	2.9	2.0	+0.67	0.4	0.6
	Surface water or rainwater	+0.23	0.7	0.4	-0.37	0.5	0.3	+0.31	2.4	1.2	+1.00	0.9	1.6
	Tube well or borehole	-0.37	2.0	2.0	-0.55	1.0	1.7	-0.12	1.6	1.3	-0.48	0.9	2.4
	Unprotected well or spring	+0.36	0.7	0.4	-0.06	0.3	0.1	-0.11	1.1	0.7	+0.91	2.3	1.6
<i>Sanitation</i>	Missing sanitation category	-0.61	1.6	1.1	-0.82	1.1	1.1	-0.17	1.6	1.0	-0.42	0.5	0.5
	Open defecation	-0.18	1.8	0.8	-0.47	1.8	1.4	+0.27	2.4	1.0	+0.06	0.9	0.9
	Other sanitation category	-0.11	0.1	0.3	-0.29	0.1	0.1	+0.20	0.2	0.3	-0.23	0.1	0.2
	Household composition												
<i>Household composition</i>	Any child under age five	+0.03	0.3	0.1	+0.04	0.3	0.1	-0.07	1.7	0.3	-0.03	0.3	0.2
	Boy age fifteen or younger	-0.18	0.0	0.1	-0.12	0.0	0.1	-0.32	0.1	0.5	-0.50	0.0	0.2
	Girl under age fifteen	-0.09	0.1	0.1	-0.36	0.1	0.1	-0.05	0.2	0.3	-0.15	0.1	0.1
<i>Source-water E. coli</i>	Source-water E. coli decile 5	+0.22	0.1	0.1	+0.04	0.1	0.1	+0.32	0.7	0.6	+0.39	0.1	0.2
	Source-water E. coli decile 6	-0.30	0.2	0.1	-0.11	0.1	0.1	+0.32	0.7	0.6	+0.57	0.2	0.3
	Source-water E. coli decile 7	+0.40	0.2	0.1	+0.27	0.2	0.1	+0.34	1.4	0.6	+0.69	0.2	0.4
	Source-water E. coli decile 8	+0.57	0.9	0.3	+0.28	0.6	0.1	+0.68	5.3	1.8	+0.83	1.4	1.0
	Country fixed effects												
<i>Country fixed effects</i>	Bangladesh	+0.26	1.5	1.9	-0.11	1.0	1.0	-2.05	1.6	3.5	+0.94	0.9	1.9
	Benin	-0.73	1.2	2.0	-8.38	0.7	1.2	-0.58	0.9	1.2	-0.26	0.8	2.2
	Central African Republic	-0.74	0.2	0.6	-7.16	0.1	0.5	-0.28	0.6	1.4	-1.12	0.3	0.7
	Chad	-0.43	0.4	0.9	-2.30	0.3	0.7	-0.38	1.5	3.0	+0.47	0.3	0.4
	DR Congo	-1.45	1.5	2.9	-1.05	0.6	1.1	-1.08	0.9	1.2	-2.64	1.1	2.8
	Dominican Republic	+1.47	0.4	0.6	+0.68	0.3	0.6	+1.99	5.5	5.1	+1.03	0.1	0.2
	Fiji	+3.10	5.7	10.7	+4.09	9.6	21.7	-0.06	0.2	0.6	+3.02	0.5	2.8
	Gambia	+0.95	1.0	1.6	-2.80	0.3	0.7	-1.33	0.2	1.2	+3.07	4.4	5.5
	Ghana	-0.60	1.0	1.8	-1.10	0.5	1.0	-1.30	0.7	1.6	+0.43	0.3	0.7
	Guinea Bissau	+4.11	16.9	13.9	-0.10	0.0	0.0	+1.44	4.9	8.0	+6.09	74.2	50.9
	Guyana	+1.22	0.2	0.7	+0.96	0.1	0.2	+1.47	9.3	9.1	<u>+1.54</u>	0.3	0.4
	Honduras	+1.91	1.6	2.5	+1.56	0.3	0.4	+1.93	21.7	12.2	<u>+2.61</u>	1.0	3.9
	Lao PDR	+2.40	3.6	6.1	+3.87	6.3	9.3	-3.70	0.8	2.0	+1.23	0.1	0.4
	Lesotho	-0.31	0.4	0.9	+1.08	0.1	0.5	-8.15	0.4	1.6	-1.18	0.3	1.0
	Madagascar	+0.20	0.5	0.5	+1.88	0.5	2.4	-1.99	1.1	2.7	-0.59	0.7	2.5
	Malawi	-0.07	0.6	0.6	+0.24	0.3	0.3	0.20	1.1	2.7	0.23	0.5	0.7
	Mongolia	+4.02	18.1	18.4	+5.51	34.3	24.8	-2.96	0.3	1.0	+1.10	0.1	0.6
	Nepal	+0.96	0.5	1.0	+0.94	0.2	0.4	-3.50	0.8	2.2	+1.94	0.3	1.1
	Sierra Leone	-0.36	0.4	0.6	-3.41	0.3	0.7	-0.60	0.5	1.3	+0.78	0.4	0.4
	Suriname	+1.12	0.2	0.8	+1.51	0.1	0.4	-1.88	0.3	1.3	+2.27	0.4	1.3
	Togo	+0.22	0.1	0.3	+1.18	0.1	0.4	+0.33	1.0	2.7	+0.72	0.1	0.0
	Trinidad and Tobago	+1.51	0.2	0.6	+1.79	0.1	0.4	+0.03	0.3	1.2	+0.89	0.1	0.4
	Viet Nam	+4.20	21.8	16.8	+5.31	24.0	19.1	-0.59	0.1	0.7	+2.99	1.0	2.6
	Zimbabwe	-1.42	1.0	1.8	-0.60	0.4	0.7	-1.51	0.5	1.5	-2.25	0.5	1.4
AUC		0.879	0.893	0.891	0.941	0.944	0.946	0.832	0.848	0.853	0.912	0.921	0.924

Notes: Each treatment is compared with no treatment. The table includes all prespecified control blocks. Categories not shown are the reference categories: poorest wealth quintile, rural residence, flush toilet, the first observed water-source and source-water E. coli categories, and the omitted country fixed-effect category. Country fixed effects are included in estimation and displayed for transparency. LASSO reports penalised logistic coefficients; a positive coefficient indicates higher treatment-adoption odds and a negative coefficient indicates lower odds, relative to the reference category. Zero indicates that the predictor was not selected. RF/XGB report direction-neutral feature importance multiplied by 100. For each treatment and learner separately, bold identifies the most predictively important variable within its block; bold and underlining identify the most predictively important variable overall. Thus, the highlighted predictor can differ across LASSO, RF, and XGB.